MAD LAB QUESTIONS

Model Question Bank (Comprehensive Syllabus UI Component Coverage)

Exam Blueprint Note: Every question below assesses the core syllabus knowledge base—Layouts, Input Validation, SharedPreferences, Menus, ListView Adapters, Explicit/Implicit Intents, Alert Dialogs, and Toasts—while exhaustively incorporating specialized UI widgets (RadioButtons, CheckBoxes, ProgressBars, Spinners, AutoCompleteTextViews, TableLayouts, and GridLayouts). All database/SQLite operations are excluded.

1. Multi-Tier Fitness Assessment Console

Design and develop an Android application across two activities. The first activity must utilize a structured **TableLayout** containing an **AutoCompleteTextView** for selecting workout routines, a **RadioGroup** with **RadioButtons** for intensity levels, and multiple medical warning **CheckBoxes**. Implement real-time input validation on the text fields, alongside an Options Menu containing a "Reset Form" action. Upon clicking a "Save Evaluation" button, prompt the user with an Alert Dialog; once confirmed, store the profile configuration metrics inside Shared Preferences and route to the second activity via an Explicit Intent. The second activity must extract these parameters and populate a tailored array of specialized exercise plans into a scrollable **ListView** using an Android Adapter. Long-pressing any routine layout must reveal a Context Menu item that fires an Implicit Intent launching external map navigation pinpointing the nearest sports facility coordinates.

2. Smart Vehicle Fleet Broker

Design and develop a vehicle dispatching Android application featuring two activities utilizing an activity result lifecycle pattern. The primary dashboard activity must showcase standard travel stats, alongside an Options Menu containing a "Book Custom Ride" action. Selecting this action must execute an Explicit Intent launching the second activity expecting a returned result. The second activity must implement a clean **GridLayout** framework embedding an **EditText** field for destination entry (with runtime text validation), a vehicle classification **Spinner**, and a horizontal **ProgressBar** indicating driver availability lookup. Tapping "Confirm Booking" must pop up an Alert Dialog confirmation; upon completion, bundle the confirmed parameters back to the parent dashboard activity as a strict callback result. The primary activity must catch this bundle, persist the configuration state via Shared Preferences, and issue a dynamic validation Toast notification.

3. Professional Flight Booking & Cargo Gateway

Design and develop a dynamic cargo booking Android utility configured across a multi-tab structural layout. The primary configuration terminal activity must arrange travel booking workflows using target destination inputs via a **Spinner**, optional service category **CheckBoxes**, and a passenger count input field enforcing strict numeric range validation. An Options Menu must allow clear operations on all entry slots. Clicking the layout's checkout sequence must present a confirmation Alert Dialog. If accepted, write the flight configuration variables into Shared Preferences, explicitly passing bundled details forward to the itinerary monitoring activity. This secondary activity must map the active flight legs onto an adapter-linked **ListView**. Long-pressing a row must activate a Context Menu configured with an Implicit Intent to render high-resolution terminal blueprints on the device's external web browser.

4. E-Learning Registration & Progress Monitor

Design and develop an educational application demonstrating programmatic view customization. The first activity acts as an intake console, prompting student credentials using an **AutoCompleteTextView** matched with department selection **RadioButtons**, ensuring strict non-empty field validation loops. Tapping "Build Student Profile" records the persistent student identity inside Shared Preferences before firing an Explicit Intent to launch the second activity. The second activity must programmatically inflate custom sub-view UI headers onto the root layout, displaying academic course milestones inside an adapter-mapped **ListView** while tracking module graduation thresholds using a dynamic **ProgressBar**. Clicking a course row generates an Alert Dialog, whereas long-pressing a row reveals a Context Menu that fires an Implicit Intent to open the phone dialer pre-loaded with an academic advisor's hotline.

5. Tele-Health Diagnostic Portal

Design and develop a clinical assessment platform. The main entry screen activity organizes symptom gathering metrics inside an analytical **TableLayout**, collecting patient names via an **EditText** box and past condition histories using distinct **CheckBox** rows. An administrative Options Menu must provide a simple "Wipe Entries" trigger to clear the table cells. Validating and submitting the form writes the diagnostic token parameters to Shared Preferences and kicks off an Explicit Intent to display the summary activity. The second activity extracts the information to render matching specialist consulting hours inside an adapter-linked **ListView**. Short-clicking an item displays an interactive Alert Dialog that updates a dynamic Toast message upon user confirmation, while long-clicking an item activates a Context Menu launching the device email client via an Implicit Intent to send a referral message.

6. Retail Point-of-Sale Checkout Hub

Design and develop a point-of-sale invoicing utility. The first activity hosts a structural invoice construction deck built around a **GridLayout** manager. It must include product catalog selection **Spinners**, numeric item count **EditText** controls under strict boundary constraints, and checkout parameter **RadioButtons**. Selecting "Process Transaction" from the primary Options Menu saves transactional hash logs directly into Shared Preferences and routes users explicitly to the breakdown activity. The second activity dynamically maps invoice rows into a structured **ListView** backed by an Android Adapter. Long-pressing an individual line item invoice component opens a Context Menu that invokes an Implicit Intent routing directly to an external global package shipping map application.

7. Corporate Recruitment Portal & Skills Matrix

Design and develop an application for engineering recruitment. The first activity renders a structured application layout mapping a candidate's profile through a specialized skills criteria grid of multi-select **CheckBoxes** and a primary language track **RadioGroup**. Clicking "Submit Candidacy" executes field validation and handles an Alert Dialog layer evaluating eligibility constraints. Upon authorization, profile flags are written to Shared Preferences before an Explicit Intent instantiates the recruitment dashboard activity. This second activity formats matching job openings inside an adapter-backed **ListView**. Tapping an opening presents a quick registration Toast, while long-pressing the line item exposes a Context Menu triggering an Implicit Intent browser redirect to download corporate employee benefits files.

8. Culinary Order & Home Delivery Logistics

Design and develop a restaurant order management console. Activity 1 captures target delivery locations using an **AutoCompleteTextView**, handles ordering criteria via a food category selector **Spinner**, and enforces runtime non-empty input field validation checks. Submitting the order fires an Alert Dialog validation review prompt, stores order indices within Shared Preferences, and transfers state explicitly to Activity 2. Activity 2 parses the intent data to render tracking logs into a custom adapter-linked **ListView**, while updating simulated fulfillment status timelines using an interactive horizontal **ProgressBar** component. Long-pressing items invokes a Context Menu item configured with an Implicit Intent to open external maps showing real-time courier paths.

9. Academic Project Repository & Archive

Design and develop an academic thesis archiving index. The first activity structures project metadata collection through a strict **TableLayout**, which embeds research title validation text inputs and department sorting **RadioButtons**. An administrative Options Menu features a single "Clear Configurations" selection to safely flush input structures. Saving records writes the index variables to Shared Preferences before executing an Explicit Intent to reveal the archive terminal screen. The second activity maps corresponding project arrays directly into an adapter-driven **ListView**. Clicking any file row launches an Alert Dialog verifying file opening actions, while long-pressing a record activates a Context Menu that fires an Implicit Intent passing project bibliography hyperlinks to the system web browser app.

10. Smart Hotel Room Management Deck

Design and develop a smart room climate and concierge configuration console. The primary panel activity outlines appliance toggles using a balanced grid layout matrix featuring lighting toggles, air conditioning control **RadioButtons**, and amenity request **CheckBoxes**. Validating the panel layout inputs registers the preference values inside Shared Preferences and executes an Explicit Intent launching the room services dashboard activity. This second activity dynamically parses the environment metrics to greet the guest and maps operational active room requests inside a scrollable **ListView** via an Android Adapter. Short-clicking item elements fires an informative confirmation Toast, whereas long-pressing row indices reveals a Context Menu executing an Implicit Intent web request detailing external hotel spa coordinates.